BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Project Review and Assessment Feedback

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. What is the purpose of testing a program against expected outputs?

2. What can unsafe SQL input let an attacker change?

A) The CPU's clock speed
B) An optical disc's capacity
C) A database query
D) A cable's length

3. Can an array be passed to or returned from a subprogram?

4. A school updates its anti-malware. "Anti-malware can detect every future threat." Correct this. Why are other controls needed?

Today's objectives

Improve programs using test results and feedback. Apply this to another software scenario.

Key vocabulary - Read, cover, recall

Test evidence

A table should record test input, expected result, actual result and outcome.

Fault diagnosis

Compare expectation and observed behaviour to identify a specific error.

Refinement

Change the faulty instruction and run the affected test again.

Feedback

Use genuine tester or teacher comments to choose improvements, not guessed marks.

Write one key term and its meaning in your own words:

Do Now	I Do	We Do	You Do	Review & Improve
Recall previous learning	Read new knowledge	Practise with guidance	Work independently	Check my understanding

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Learn and practise

New learning - Key ideas

Test evidence	A table should record test input, expected result, actual result and outcome. Consider this in a project correction.
Fault diagnosis	Compare expectation and observed behaviour to identify a specific error. Consider this in a quiz total review.
Refinement	Change the faulty instruction and run the affected test again. Consider this in a ticket app test failure.
Feedback	Use genuine tester or teacher comments to choose improvements, not guessed marks. Consider this in a club database feedback.

I Do - One worked example

A project correction needs test evidence and fault diagnosis. Explain why. [4 marks]

1. Identify the inputs, constraints and expected result.

2. Test evidence [1]; A table should record test input, expected result, actual result and outcome. [1].

3. Fault diagnosis [1]; Compare expectation and observed behaviour to identify a specific error. [1].

4. Give a specific point and a linked reason for each feature.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. In a quiz total review, explain the purpose of test evidence. [2 marks]

2. A ticket app test failure needs fault diagnosis. Explain its role. [2 marks]

3. A club database feedback uses refinement and feedback. Explain both. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Correct the calculation so it prints the price of three GBP 6 items.

```
unitPrice = 6
quantity = 3
total = unitPrice + quantity
print(total)
```

BLACK PEN: predict or complete the code; test, then improve.

You Do - Independent exam questions

Q2. State TWO concepts relevant to this topic. [2 marks]

Q3. Explain how fault diagnosis and refinement help in a café receipt bug. [4 marks]

Q4. Explain the roles of test evidence, refinement and feedback in a pupil scoring app. [6 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. A table should record test input, expected result, actual result and outcome. ☐ T ☐ F
2. A table should record test input, expected result, actual result and outcome ☐ T ☐ F
3. Compare expectation and observed behaviour to identify a specific error ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 To compare its actual behaviour with the requirements and identify faults. ☐

Q2 C - A database query ☐

Q3 Yes. A subprogram can receive an array; a function can return an array. ☐

Q4 Anti-malware detects, quarantines or removes malware. Updates help it recognise threats. It cannot stop every attack, so other controls are needed. ☐

Code First - Model

```
unitPrice = 6
quantity = 3
total = unitPrice * quantity
print(total)
```

Output: 18

We Do - Check the model

1. Test evidence [1]; A table should record test input, expected result, actual result and outcome. [1]. ☐

2. Fault diagnosis [1]; Compare expectation and observed behaviour to identify a specific error. [1]. ☐

3. Refinement [1]; Change the faulty instruction and run the affected test again. [1]. Feedback [1]; Use genuine tester or teacher comments to choose improvements, not guessed marks. [1]. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 Test evidence [1]; Refinement [1]. ☐

Q3 Fault diagnosis [1]; Compare expectation and observed behaviour to identify a specific error. [1]. Refinement [1]; Change the faulty instruction and run the affected test again. [1]. ☐

Q4 Test evidence [1]; A table should record test input, expected result, actual result and outcome. [1]. Refinement [1]; Change the faulty instruction and run the affected test again. [1]. Feedback [1]; Use genuine tester or teacher comments to choose improvements, not guessed marks. [1]. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: